Project Proposal: Urban Transport Classifier

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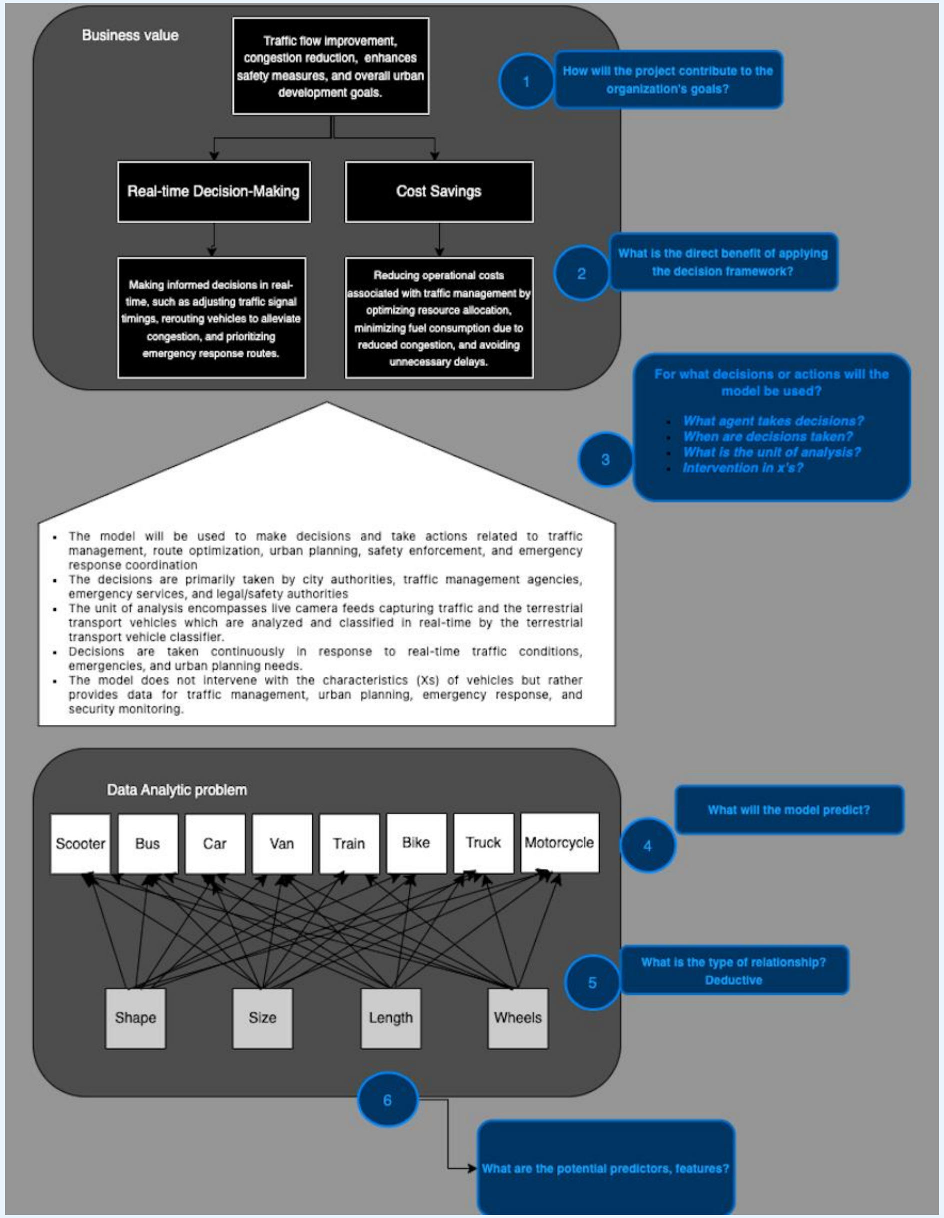
Project Pitch

Business Understanding

- Market Research: Traffic congestion in metropolitan areas poses significant challenges, including delays, safety hazards, and economic losses, highlighting the need for innovative solutions to enhance traffic management and urban mobility.
- Main Stakeholder: City Authorities
  - Real-time data for informed decision-making.
  - Tools to optimize traffic flow, reduce congestion, and enhance safety.
  - Easy integration into existing infrastructure.
  - Cost-effective and scalable solutions.
  - Collaboration for innovative solutions aligned with urban mobility goals.
- Idea: Integrating a terrestrial transport classifier into camera systems allows for real-time analysis of live feeds, automatically identifying and classifying various types of urban transport vehicles.



Problem Overview

Deep Learning

- **My image classification problem:**
  - Classification of various types of terrestrial transport commonly used in urban areas.
  - 8 different classes
  - Scooter, Bike, Motorbike, Car, Bus, Van, Truck and Train
  - 150 handpicked images per class
  - **Random Guess Accuracy** $=\frac{1}{n} = \frac{1}{8} = 12.5\%$
  - **Average Human-Level Performance Accuracy** = $99\% + 99\% + 90\% + 94\% + 93\% + 94\% + 99\% + 90\% + 98\% + 99\% = 95.5\%$
  - **Basic Multilayer Perceptron Baseline** ranged from 34.99% to 48.33%

Model Overview

Deep Learning – Data Preprocessing

Iteration 1 & 3

1. Loading the images
2. Resizing the images
   - 256 x 256 pixels
3. Label encoding the images
4. One-hot encoding the images
5. Splitting the dataset into training, validation, and test sets
   - 70%/20%/10%

Iteration 2 & 4

1. Loading the images
2. Resizing the images
   - 256 x 256 pixels
3. Augumenting the images
   - Each image once
4. Label encoding the images
5. One-hot encoding the images
6. Splitting the dataset into training, validation, and test sets
   - 70%/20%/10%

Model Overview

Deep Learning – Building the Model

- Experimentation with various architectures, hyperparameters, layer configurations, and activation functions…
- Conv2D, Dropout, Dense, BatchNorm, and MaxPooling2D, Flatten…
- More complex architectures hurt the model performance
- Validation accuracy ranging from 12% to 60%
- Chosen model - most balanced results, the lowest loss, minimal overfitting, and relatively high accuracy

```python
# Instantiate the model
V1_model = Sequential([
    Conv2D(32, (3, 3), activation='relu',
           input_shape=(256, 256, 3),
           padding='same', strides=(2, 2)),
    MaxPooling2D((2, 2)),
    Conv2D(64, (3, 3), activation='relu',
           padding='same', strides=(2, 2)),
    MaxPooling2D((2, 2)),
    Flatten(),
    Dropout(0.5),
    Dense(128, activation='relu'),
    Dense(8, activation='softmax')
])
```

- Iteration 1 **– Plain CNN**
- Iteration 2 **– Plain CNN with Augumented Data**
- Iteration 3 **– VGG16**
- Iteration 4 **– VGG16 with Augumented Data**

Model Overview

Deep Learning – Model Architectures

Iteration 1 & 2

Layer (type)	Output Shape	Param #
conv2d_23 (Conv2D)	(None, 128, 128, 32)	896
max_pooling2d_23 (MaxPooling2D)	(None, 64, 64, 32)	0
conv2d_24 (Conv2D)	(None, 32, 32, 64)	18496
max_pooling2d_24 (MaxPooling2D)	(None, 16, 16, 64)	0
flatten_8 (Flatten)	(None, 16384)	0
dropout_6 (Dropout)	(None, 16384)	0
dense_18 (Dense)	(None, 128)	2097280
dense_19 (Dense)	(None, 8)	1032

Total params: 2117704 (8.08 MB)
Trainable params: 2117704 (8.08 MB)
Non-trainable params: 0 (0.00 Byte)

Iteration 3 & 4

Layer (type)	Output Shape	Param #
vgg16 (Functional)	(None, 8, 8, 512)	14714688
flatten_17 (Flatten)	(None, 32768)	0
dropout_13 (Dropout)	(None, 32768)	0
dense_38 (Dense)	(None, 128)	4194432
dense_39 (Dense)	(None, 8)	1032

Total params: 18910152 (72.14 MB)
Trainable params: 4195464 (16.00 MB)
Non-trainable params: 14714688 (56.13 MB)

Model Overview

Deep Learning – Training the Model

Compiling the model

- optimizer → Adam
  - adaptive learning rate
- loss calculation → categorical_crossentropy
  - multi-class classification problem
- metrics → accuracy
  - monitoring model performance

EarlyStopping callback

- prevent overfitting and efficiently halt training in case of no progress
  - monitor → validation loss
  - patience → 2 epochs
  - restore_best_weights → True

Training the model

- fit method
  - epochs → 20
  - batch_size → 32
  - validation data → validation set for given Iteration
  - callback → EarlyStopping callback

Model Performance

Deep Learning – Iteration 1

- **Test Loss:** 0.951900064945221
- **Test Accuracy:** 0.6916666626930237

- **Best Performing Class:** 6 (motorcycle)
- **Worst Performing Class:** 5 (truck)

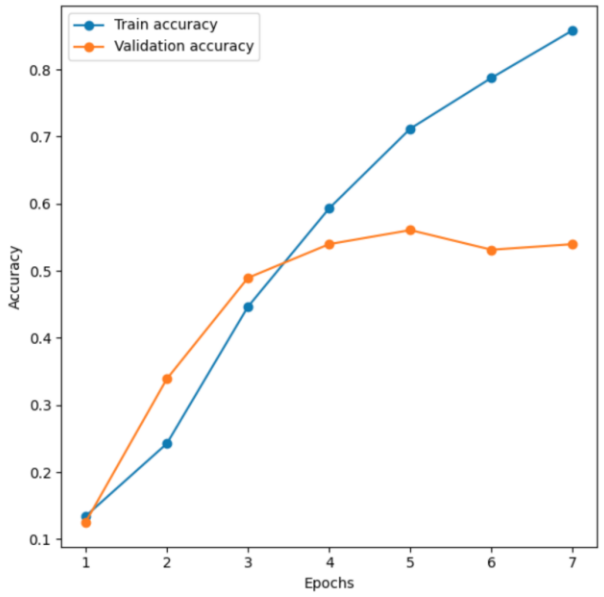
- **Weighted Average Precision:** 0.71
- **Weighted Average Recall:** 0.69
- **Weighted Average F1-score:** 0.69

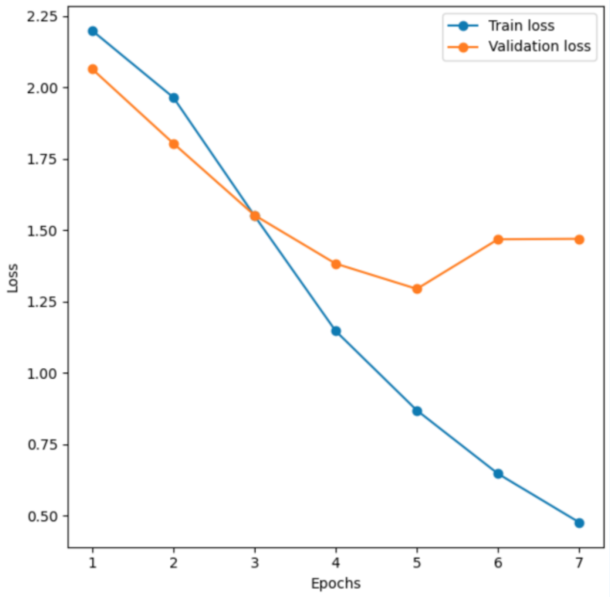
```
Confusion matrix:
[[ 6  3  0  0  1  0  1  0]
 [ 0 10  0  1  1  2  1  0]
 [ 0  0 14  0  0  0  0  2]
 [ 0  0  3 12  0  0  0  0]
 [ 1  3  0  1  7  0  0  0]
 [ 0  1  3  1  1  8  1  0]
 [ 1  1  2  1  0  1 16  0]
 [ 0  0  1  2  0  1  0 10]]
```

```
Classification report:
              precision    recall  f1-score   support

           0       0.75      0.55      0.63        11
           1       0.56      0.67      0.61        15
           2       0.61      0.88      0.72        16
           3       0.67      0.80      0.73        15
           4       0.70      0.58      0.64        12
           5       0.67      0.53      0.59        15
           6       0.84      0.73      0.78        22
           7       0.83      0.71      0.77        14

    accuracy                           0.69       120
   macro avg       0.70      0.68      0.68       120
weighted avg       0.71      0.69      0.69       120
```





Model Performance

Deep Learning – Iteration 2

- **Test Loss:** 1.269450306892395
- **Test Accuracy:** 0.5564853549003601

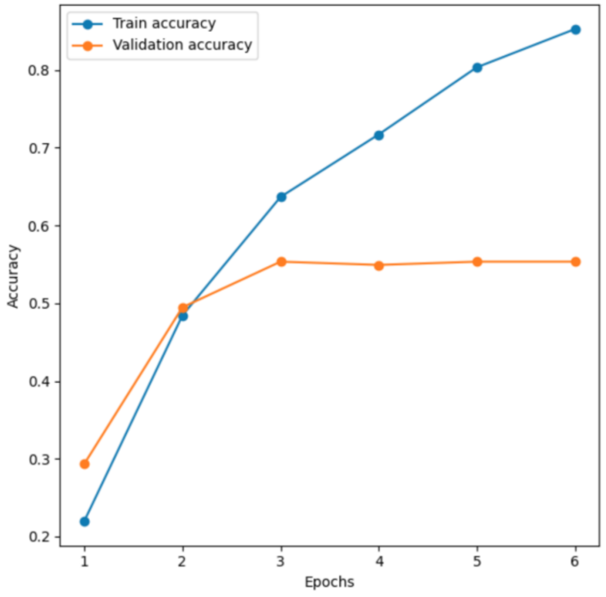
- **Best Performing Class:** 2 (bicycle)
- **Worst Performing Class:** 5 (truck)

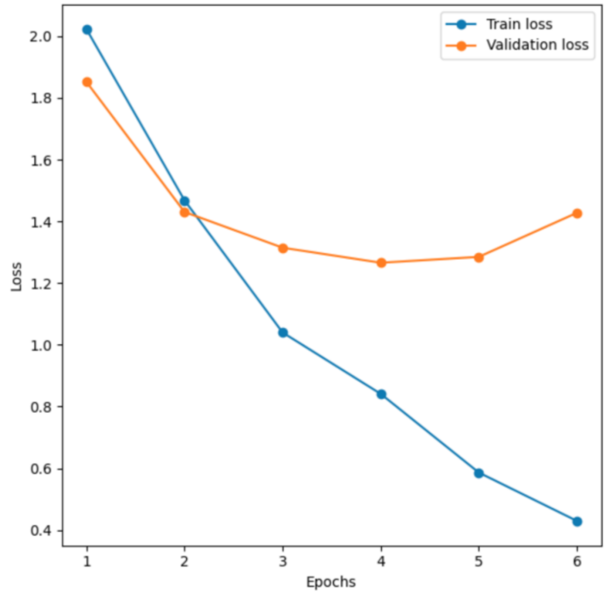
- **Weighted Average Precision:** 0.59
- **Weighted Average Recall:** 0.56
- **Weighted Average F1-score:** 0.54

```
Confusion matrix:
[[18  0  0  1  9  0  0  2]
 [ 3 21  0  4  6  2  1  0]
 [ 0  0 19  5  0  1  2  8]
 [ 2  1  0 15  3  0  3  2]
 [ 4  1  0  1 15  1  2  3]
 [ 4 10  2  1  4  5  4  2]
 [ 0  0  0  0  3  0 15  3]
 [ 2  1  1  0  1  0  1 25]]
```

Classification report:

	precision	recall	f1-score	support
0	0.55	0.60	0.57	30
1	0.62	0.57	0.59	37
2	0.86	0.54	0.67	35
3	0.56	0.58	0.57	26
4	0.37	0.56	0.44	27
5	0.56	0.16	0.24	32
6	0.54	0.71	0.61	21
7	0.56	0.81	0.66	31
accuracy			0.56	239
macro avg	0.57	0.56	0.54	239
weighted avg	0.59	0.56	0.54	239





Model Performance

Deep Learning – Iteration 3

- **Test Loss:** 0.1643863320350647
- **Test Accuracy:** 0.9416666626930237

- **Best Performing Class:** 6 (motorcycle)
- **Worst Performing Class:** 5 (truck)

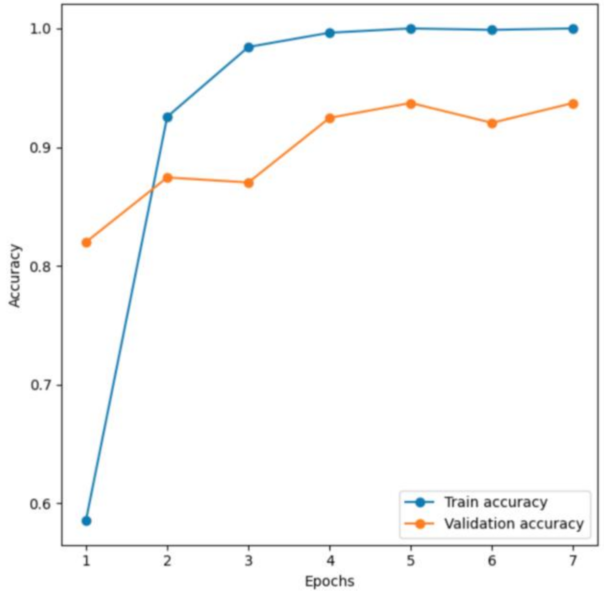
- **Weighted Average Precision:** 0.94
- **Weighted Average Recall:** 0.94
- **Weighted Average F1-score:** 0.94

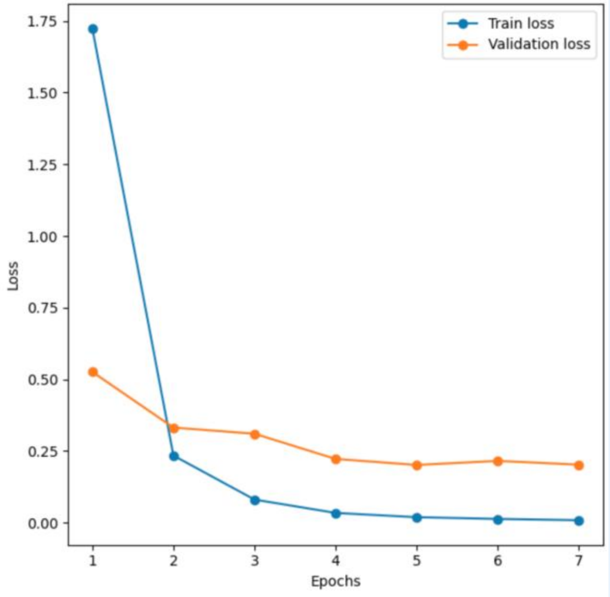
```
Confusion matrix:
[[11  0  0  0  0  0  0  0]
 [ 0 13  0  0  0  2  0  0]
 [ 0  0 15  0  0  0  0  1]
 [ 0  0  0 15  0  0  0  0]
 [ 1  0  0  0 11  0  0  0]
 [ 0  0  0  1  1 13  0  0]
 [ 0  0  1  0  0  0 21  0]
 [ 0  0  0  0  0  0  0 14]]
```

```
Classification report:
              precision    recall  f1-score   support

           0       0.92      1.00      0.96        11
           1       1.00      0.87      0.93        15
           2       0.94      0.94      0.94        16
           3       0.94      1.00      0.97        15
           4       0.92      0.92      0.92        12
           5       0.87      0.87      0.87        15
           6       1.00      0.95      0.98        22
           7       0.93      1.00      0.97        14

    accuracy                           0.94       120
   macro avg       0.94      0.94      0.94       120
weighted avg       0.94      0.94      0.94       120
```





Model Performance

Deep Learning – Iteration 4

- **Test Loss:** 0.23254236578941345
- **Test Accuracy:** 0.9246861934661865

- **Best Performing Class:** 6 (motorcycle)
- **Worst Performing Class:** 4 (van)

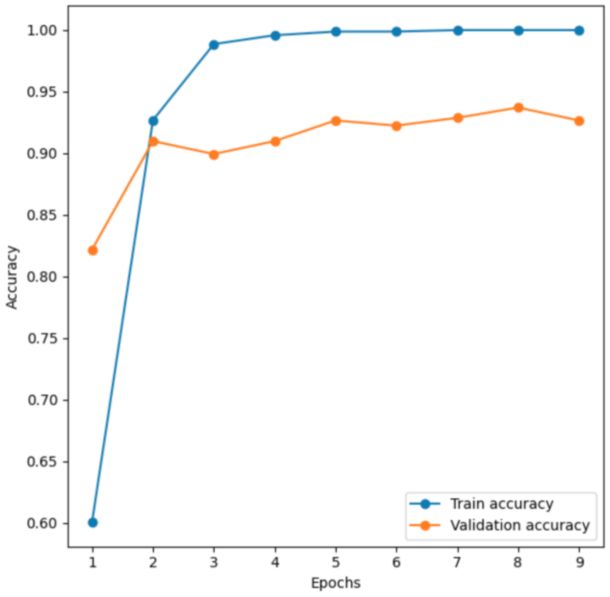
- **Weighted Average Precision:** 0.93
- **Weighted Average Recall:** 0.92
- **Weighted Average F1-score:** 0.93

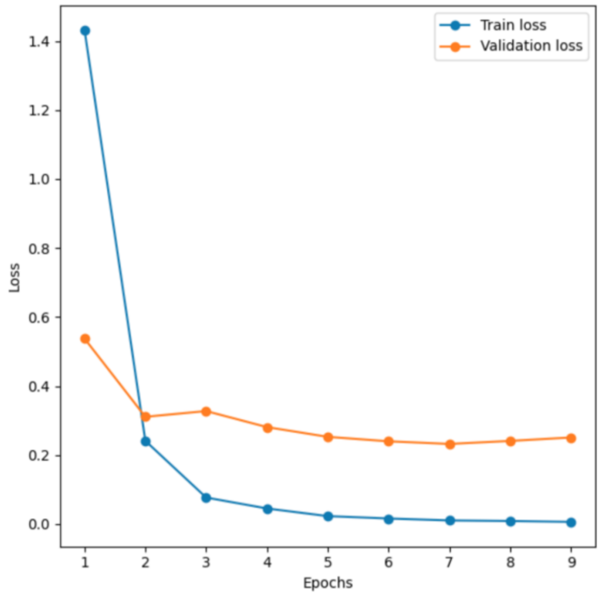
```
Confusion matrix:
[[27  0  0  0  3  0  0  0]
 [ 0 35  0  0  1  1  0  0]
 [ 0  0 34  0  0  0  0  1]
 [ 0  1  0 23  0  2  0  0]
 [ 2  0  0  0 22  3  0  0]
 [ 0  2  0  1  0 29  0  0]
 [ 0  0  0  0  0  0 21  0]
 [ 0  0  0  0  0  1  0 30]]
```

```
Classification report:
              precision    recall  f1-score   support

           0       0.93      0.90      0.92        30
           1       0.92      0.95      0.93        37
           2       1.00      0.97      0.99        35
           3       0.96      0.88      0.92        26
           4       0.85      0.81      0.83        27
           5       0.81      0.91      0.85        32
           6       1.00      1.00      1.00        21
           7       0.97      0.97      0.97        31

    accuracy                           0.92       239
   macro avg       0.93      0.92      0.93       239
weighted avg       0.93      0.92      0.93       239
```





Model Interpretability

Responsible AI

- **Accuracy & Interpretability Trade-Off:**
    - Achieving a **high level of accuracy** holds greater importance than ensuring interpretability due to its crucial role in enabling **effective real-time decision-making**, adapting to the **complexities of urban traffic environments**, and **reducing manual intervention** for **improved operational efficiency**.

User Study

Human-Centered AI

Think–Aloud Study

Three Main Findings:

- Highlighting key elements for better visibility
  - *“Skip” option (Welcome page)*
  - *“Predictive Vehicle” (Analysis Page)*
- Clarifying visual elements (Home Page) by clear text and distinct visuals
- Enhancing interactivity of non-functional elements
  - *Hamburger menu (Camera page) → Dropdown*

User Study

Human-Centered AI –

Study Elements

- Version A: Dropdown menu connecting all locations and individual cameras
- Version B: Individual interactive buttons for each location, tapping on individual cameras

Hypothesis

- Users will enjoy B version more
  - More intuitive, interactive and functional
  - Better flow
  - Simpler filtering and navigating through cameras and analysis page

Results

- **Enjoyment Factor:** Version B.
- **Page Connectivity and Navigation:** Version B
- **Comprehension and Usefulness:** Version A = Version B

Demo

Human-Centered AI

RoadReader Demo Video

"**RoadReader** - the ultimate solution for enhanced road safety and monitoring. RoadReader seamlessly connects the cameras purchased from our company to your account, offering a centralized platform for comprehensive surveillance. What sets RoadReader apart is its integrated urban transport classifier, providing users with real-time insights and classification of vehicles within the monitored areas."

Thank you!

Any questions?